Investigation of the Effects of Ce on OER performance of Ir-Ce mixed oxides

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The acidic oxygen evolution reaction (OER) is central to the proton-exchange membrane (PEM) water electrolysis, the next-generation technology along the electricity-to-hydrogen transition. Prior studies have revealed that iridium oxide (IrO_2), a scarce and precious catalyst, is among the few materials capable of enduring the demanding conditions of high potential, low pH, and high oxygen concentration at the anode. However, it falls short of meeting the performance and stability demands of GW-scale PEM electrolyzers[1]. Our previous research[2] showed that at potentials exceeding 1.6 V vs. reversible hydrogen electrode (RHE), the OER degradation mechanism of rutile IrO_2 changes as lattice oxygen interacts with the active sites, resulting in a deactivation[3].

To address the limitations associated with the use of iridium and improve catalytic performance, we explored the use of mixed Ir-based oxides ($Ir_xM_{1-x}O_2$)[4]. These materials can modify the adsorption dynamics and subsequent reactions of various adsorbates on the surface, thereby enhancing acidic OER catalytic activity[5].

We synthesized $Ir_xCe_{1-x}O_2$ (x=0.2, 0.5, 0.8) using flame spray pyrolysis (FSP). The iridium loading in $Ir_{0.5}Ce_{0.5}O_2$ and $Ir_{0.8}Ce_{0.2}O_2$ is 43 wt% and 16 wt% compared to pure IrO_2, respectively. Despite the lower Ir amount, these catalysts demonstrated improved activity (lower overpotential at 10 mA cm^{-2}) and superior stability compared to rutile IrO_2 in cyclic voltammetry (CV) measurements using a rotating disk electrode. The rising intensity of the redox peaks with an increasing number of CV cycles suggests an enhancement in the oxidation state of $Ir_{0.5}Ce_{0.5}O_2$ (Fig. 1a). The shift of the CV in

the pre-OER region (0.4 V – 1.45 V vs RHE) towards in scan direction indicates an increase in activation barrier at the Ir-centers without affecting the exponential OER currents (> 1.5 V). Some first *ex situ* XAS investigations showed that the white line position of Ir in $Ir_{0.5}Ce_{0.5}O_2$ shifts to lower values by approximately 0.5 eV compared to that of rutile IrO_2 (Fig. 1b). These findings underscore the potential of mixed Ir-based oxides as effective catalysts for advancing PEM water electrolysis technology and will be deepened by in situ EXAFS measurements as on IrO_2-catalysts[2, 3].

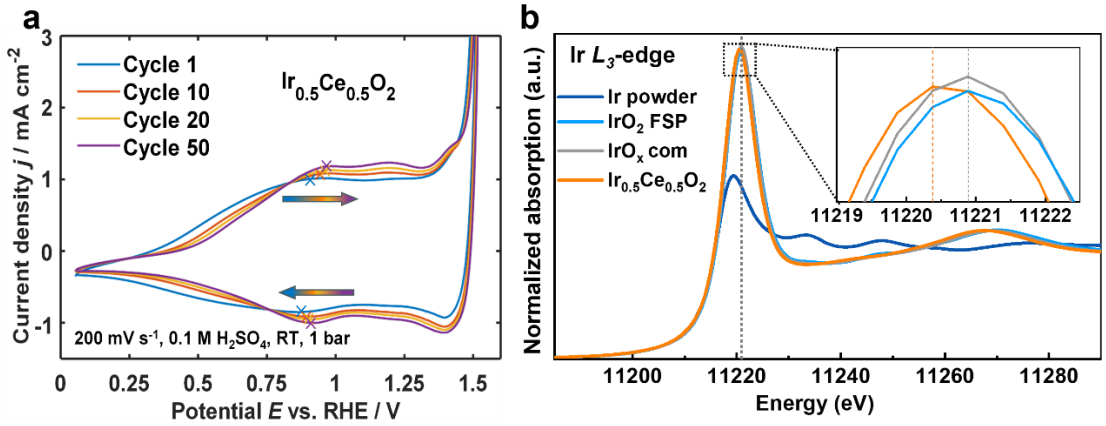

Fig.1. (a) CV curves with multiple cycles of $Ir_{0.5}Ce_{0.5}O_2$ at the scan rate of 200 mVs^{-1} (b) Ir L_3-edge XANES spectra of the Ir samples, with an inset showing an enlarged view of the white line region.

References

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